

Interview Questions

SQL Questions

1. What is SQL, and why is it essential for data professionals?
2. Explain the difference between SQL and NoSQL databases.
3. What are the primary SQL commands for data retrieval?
4. What are the main SQL data types?
5. What is a primary key, and why is it important?
6. What is a foreign key, and how is it used in SQL?
7. How do you retrieve all records from a table?
8. Explain the WHERE clause and its use in SQL queries.
9. What is the purpose of the ORDER BY clause?
10. How do you filter results based on multiple conditions using the AND and OR operators?
11. What is the LIMIT clause, and when is it used?
12. Explain the GROUP BY clause and its role in aggregation.
13. What is the HAVING clause, and how does it differ from the WHERE clause?
14. What is a SQL subquery, and how is it used?
15. How do you use the DISTINCT keyword in SQL?
16. What is the purpose of the CONCAT() function?
17. What is an INNER JOIN, and how does it work?
18. Explain LEFT JOIN and RIGHT JOIN with examples.
19. What is a self-join, and why would you use it?
20. Describe the differences between primary keys and unique constraints.
21. How do you establish a many-to-many relationship in SQL?
22. Explain the INSERT statement and how it is used.
23. What is an SQL transaction, and why is it important?
24. How can you update data in a table using SQL?
25. What is the purpose of the DELETE statement?
26. Describe the differences between DELETE and TRUNCATE.

27. Explain the concept of indexing in databases.
28. What is normalization, and why is it important?
29. How can you create and manage views in SQL?
30. What are stored procedures, and when are they useful?
31. Explain SQL injection and methods to prevent it.
32. How do you optimize SQL queries for performance?
33. What is a unique key in SQL?
34. What is the difference between UNION and UNION ALL?
35. What is the difference between a table and a view?
36. What SQL operator is used for pattern matching?
37. What's a SQL aggregate function and how is it used?
38. What DDL and DML?
39. Give me a brief of DBMS, RDBMS and eventually MYSQL ?
40. Explain database relationships and mention their various types
41. Define constraints and give five examples of them. - CHECK,DEFAULT,UNIQUE,PRIMARY KEY,FOREIGN KEY
42. Mention a major distinction among the Non-Cluster Index and Cluster Index?
43. What do you understand about user-defined functions?
44. What is Cross-Join?
45. What is a Data Warehouse?
46. What are scalar and aggregate functions?
47. What is ETL in SQL ?
48. What are Nested Triggers?
49. What will be the SQL query to get the employee full names and replace the space with '-'.
50. Explain the main features of a relational database
51. What is an ACID property in SQL?
52. What is the difference between SQL and MySQL?
53. What are SQL comments?

